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Quantum-safe blockchain-assisted data encryption protocol for internet of things networks

Sunil Prajapat[1] · Neeraj Kumar[2] · Ashok Kumar Das[3,4] · Pankaj Kumar[1] · Rifaqat Ali[2]

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Abstract

The Internet of Things (IoT) is rapidly growing, providing convenience in daily life. However, ensuring the security of IoT systems is crucial, especially as they generate vast amounts of data, often containing personal and private information. To address these issues, we propose a quantum-safe blockchain attested encryption scheme that uses lattice-based encryption with blockchain technology. This consensus process removes reliance on third-party central service providers and uses consensus nodes as proxy service nodes to encrypt data and aggregate translated ciphertext. This approach mitigates the limitations of using a decentralized network for secure storage and dissemination of private encrypted data. Moreover, the security analysis demonstrates security against attacks from polynomial-time adversaries and probabilistic polynomial-time adversaries. Additionally, the performance evaluation shows that the proposed scheme outperforms existing ones in terms of computational and communication costs. The encryption and decryption costs are minimized to $T_{*1} + (T_{*2} + T_{*3} + T_{+1}) \approx 3.096$, and $T_{*6} \approx 0.010$, respectively, demonstrating its efficiency. A comparison of properties and features with other protocols highlights the proposed scheme's advantages, such as quantum resistance, multi-authority support, and collision resistance. This combination establishes the proposed scheme as a robust solution. The proposed protocol employs lattice-based encryption and blockchain consensus algorithms to facilitate IoT data exchange, making it highly efficient and secure for the transfer of information in IoT systems.

Keywords Identity-based encryption · Post-quantum security · Internet of Things (IoT) · Blockchain · Learning with error

✉ Pankaj Kumar
pkumar240183@gmail.com

Sunil Prajapat
sunilprajapat645@gmail.com

Neeraj Kumar
neerajmnnit2016@gmail.com

Ashok Kumar Das
iitkgp.akdas@gmail.com; ashok.das@iiit.ac.in

Rifaqat Ali
rifaqatali@nith.ac.in

1 Srinivasa Ramanujan Department of Mathematics, Central University of Himachal Pradesh, Dharamshala 176 206, India

2 Department of Mathematics and Scientific computing, National Institute of Technology, Hamirpur 177 005, India

3 Center for Security, Theory and Algorithmic Research International Institute of Information Technology, Hyderabad 500 032, India

4 Department of Computer Science and Engineering, College of Informatics, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul 02841, South Korea

1 Introduction

In 1999, "MIT's Automatic Identification Centre" introduced the notion of the Internet of Things (IoT) for the first time. IoT has been widely applied in several practical contexts. Previous years applications Kevin defined the IoT as a universal and standardised network that incorporates "Radio Frequency Identification (RFID)" and various other sensors [1, 2]. In 2005, the "International Telecommunication Union" (ITU) released a paper that broadened the definition of the IoT [3]. The term IoT refers to the expansion of the Internet, encompassing technologies such as RFID, sensor networks, and smart gadgets [4]. IoT technology enables the collection of data by devices, which can then be processed instantaneously. By implementing this approach, the overall efficiency of the system can be enhanced while simultaneously reducing expenses [5]. Due to its significant impact on both individuals lives and industrial production, the IoT has garnered increasing

research and interest. Specifically, as hardware technology has advanced, the associated gadgets of the IoT have undergone significant optimisation and improvement. "Wireless sensor networks (WSN)", "machine-to-machine (M2M)", and "cyber-physical systems (CPS)" are now integral components of the IoT [6–10]. The potential scope of IoT has significantly broadened in recent years.

However, as the IoT continues to advance, it is imperative that its security challenges are not overlooked. The IP protocol serves as the primary standard for connectivity, however certain security concerns persist in IoT applications involving WSN, M2M, and CPS. Hence, it is imperative to safeguard the entire deployment architecture from potential attacks by any individual. Furthermore, it is imperative to prevent any potential risks that may compromise the privacy, integrity, or confidentiality of data [11, 12]. Furthermore, due to the integration of the Internet and diverse devices, the IoT model also encompasses conventional security issues associated with computer networks. IoTs encounter unique security challenges due to their resource constraints and the sensitive nature of the data they manage, which often includes identification and location information. Additionally, these networks are characterized by open accessibility to various user devices, heightening the significance of data breaches and privacy threats. Therefore, ensuring the confidentiality, integrity, and availability of data within IoTs is paramount. Robust security measures are required to protect against unauthorized access, data breaches, and disruptions in data collection and communication. Figure 1 illustrates the communication architecture of the IoT network, its various applications, and potential security threats. Authentication, a foundational pillar of IoT security, assumes critical importance in the context of IoT. It serves as the solution to address these challenges effectively and provides a means to verify the identity and legitimacy of devices within the IoT environment, ensuring that only authorized entities can access sensitive data and participate in network operations.

The distribution of devices in the IoT is widely dispersed, and manufacturers have a limited understanding of security, resulting in many devices lacking essential security mechanisms such as encryption, authentication, and access control. Oftentimes, the process of connecting a device does not necessitate the use of a password for authentication. Furthermore, several early IoT devices continue to utilise extremely obsolete software versions, rendering them incapable of attaining remote upgrades [13, 14]. These software vulnerabilities render IoT devices very susceptible to attacks. Hence, by means of unauthorised entry or a "Distributed Denial of Service (DDoS)" assault, malevolent individuals can infiltrate our IoT infrastructure and manage our smart gadgets. Over the past year, the IoT system has experienced numerous significant network security breaches worldwide. In a separate publication, Komninos determined that any security framework for the Internet of Things (IoT) must fulfil three primary security prerequisites: confidentiality, integrity, and availability, in that order [15].

The introduction of blockchain technology occurred in 2008 when Nakamoto first presented it [16]. Blockchain is a revolutionary technology that operates on a decentralised and transparent ledger system, eliminating the need for trust and central authority. It is also distributed, meaning that the information is stored across multiple computers or nodes [17]. This technology enables a significant number of nodes to actively engage in the system and exchange information with each other within a specific time frame. To ensure the validity of transaction information, signatures are generated and data blocks are linked through cryptographic associations for verification [18]. Finally, through the consensus method [19], all nodes collectively share and keep an identical public database in their own local storage. By integrating the "Elliptic Curve Digital Signature Algorithm", distributed storage, and consensus mechanisms, blockchain technology allows users who have little trust in each other to establish and uphold consensus in a decentralised network [20–22].

There exists an increasing quantity of research in recent years focusing on the integration of IoT with blockchain technology [23]. In 2017, Li et al. presented a multi-layer secure IoT network model [24–26] by implementing blockchain technology. This architecture can mitigate the challenges associated with deploying IoT systems and offer a comprehensive solution for wide-area networking. Regrettably, it employs a centralised approach to access management. When access control queries and updates occur often, using a single centralised access control server will cause an increase in the system's workload. Meanwhile, Huh et al. proposed a system for controlling Internet of Things (IoT) devices through the utilisation of Ethereum [27]. Nevertheless, his scheme lacks a comprehensive security examination.

The concept of multi-authority in attribute-based encryption (ABE) refers to a decentralized approach where multiple attribute authorities are involved in managing and issuing access control policies and keys to users. In a multi-authority ABE scheme, each attribute authority is responsible for a subset of attributes and has the autonomy to issue keys based on those attributes without relying on a central key generation center. In the context of the scheme architecture and algorithm design, the term "multi-authority" implies that the proposed lattice-based IBE scheme involves multiple attribute authorities, each managing a distinct set of attributes. These attribute authorities collaborate to collectively enforce access control policies and manage user keys.

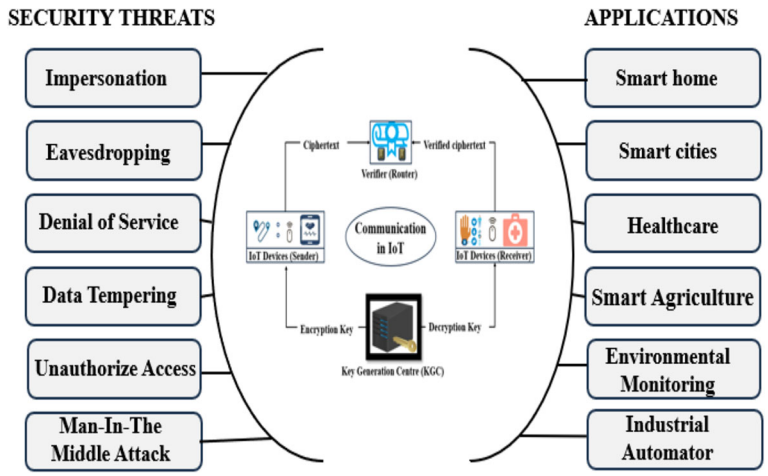


Fig. 1 Applications, threats and communication of data in IoT

Lattice-based encryption is widely recognized as one of the leading forms of encryption that is robust to quantum attacks. In 1994, Shor [28] introduced a quantum method that has the potential to solve challenging issues including discrete logarithm, elliptic curve discrete logarithm, and massive integer factorization difficulties. Consequently, all the aforementioned approaches may lack security, and other encryption solutions that are immune to quantum attacks have been suggested in recent years. Lattice-based encryption has gained interest due to its capacity to integrate functional encryption. Lattice-based encryption relies on the shortest vector problem (SVP) and related problems for its security. Furthermore, there is currently no algorithm that is capable of solving the Shortest Vector Problem (SVP) in polynomial time. The problem of solving approximate-SVP can only be solved in polynomial time [29], but the problem of solving SVP can be addressed in polynomial time.

Development on lattice-based attribute-based encryption has been conducted in recent years. In 2012, Agrawal et al. [30] introduced a Fuzzy identity-based encryption system based on lattices, and in the same year, Zhang et al. [31] enhanced it to a lattice-based ciphertext-policy attribute-based encryption scheme. In 2013, Boyen [32] introduced the first lattice-based key-policy attribute-based encryption technique. Subsequently, Zhang et al. [33] introduced the first lattice-based multi-authority attribute-based encryption system in 2015, followed by a variation proposed by Liu et al. [34] in 2018. Both Zhang et al.'s system and Liu et al.'s scheme are based on Key-Policy Attribute-Based Encryption.In 2021, Datta et al.[35] introduced the first lattice-based multi-authority attribute-based encryption technique, specifically the ciphertext-policy attribute-based encryption scheme. While some proposals for multi-authority attribute-based encryption on lattices have been made, as far as we are aware, there is currently no lattice-based multi-authority attribute-based encryption scheme that includes blockchain technology. In the current scenario, IoT applications require a quantum-safe blockchain-assisted data encryption protocol. Our protocol addresses this need by mitigating a variety of challenges related to quantum attacks and data storage problems.

1.1 Motivation and contribution

In contrast to previous research, we consider the lattice-based encryption scheme as a significant technological advancement. In this work presents a robust lattice-based Encryption scheme, which is then integrated into the access control system of the IoT using blockchain technology. It can implement precise access control and enhance the efficiency and security of accessing the IoT system. The primary contributions of this article are outlined as follows

- In this research, we introduce a robust lattice-based Encryption scheme using the "Shortest Integer Solution (SIS)" problem. It has the capability to accommodate adaptable access policies and offers user privacy protection. Furthermore, it incorporates meticulous access control in the IoT.
- By integrating the lattice-based Encryption scheme with blockchain technology, and also present a robust access control scheme for the IoT that ensures security. Within this novel framework, we employ blockchain technology for the purposes of authentication and access management. By employing blockchain's signature verification, it is possible to achieve secure and dependable identity verification for IoT devices and their communication data, hence ensuring the security of the IoT.
- We present the security analysis of the proposed scheme and a comparative performance analysis with other encryption schemes to show that our scheme is well-fit for the IoT. Although no scheme has been put

forward in the case of the IoT, the comparative study is done to demonstrate that our scheme is very cost-efficient, and suitable for the IoT.

1.2 Paper structure

The remaining structure of the paper is arranged as follows: Sect. 2 covers preliminaries, including lattices and their hard problems. Section 3 discusses the system model, syntax, and requisites for security. Section 4 presents the proposed encryption scheme with blockchain. Section 5 conducts the security analysis of the proposed scheme. Section 6 includes the efficiency assessment of the proposed work and a comparative performance study with relevant competing schemes. Finally, in Sect. 7, we conclude the work.

Table 1 Notation table

Symbol	Interpretation
P	Prime number
M, N	Positive integers
H_1, H_2	One way hash functions
T_{A_o}	Master secret key
A_o	Master public key
ID_s	Identity of sender
ID_r	Identity of receiver
B_1	Encryption key
E_1	Decryption key
CT	ciphertext
A	Attacker/adversary
v	Random vector
a, b	Noise vectors

2 Mathematical preliminaries

Assign the sets of real and integer numbers, R and Z, respectively. We represent "Z_P as addition (mod P)" with the values $\{0, 1, ..., P-1\}$. Write Z^M the collection of integer vectors. We refer to vectors as being Z_P independent if they remain linearly independent when reduced (mod P). Assume "M be a positive integer, $[M]$ is stand for $\{1, 2, ..., M\}$, and $\lceil M \rceil$, and $\lfloor M \rfloor$denote the minimum integer larger than M and the maximum integer smaller than M", respectively. Additionally, we present vectors and matrices in lowercase letters and uppercase letters, where a_j means the jth column vector of a matrix A and a_j stand for j^{th} component of a vector a, respectively. Additionally, $\tilde{A}$ represents the "Gram-Scahmidt orthogonalization of A". "$\|A\|$ and $\|a\|$ stand for the Euclidean norm of A and a". A randomized algorithm that operates in strict polynomial time is known as a "probabilistic polynomial-time (PPT)" algorithm. Table 1 provides the notations utilized in protocol.

2.1 Lattice

Assume that A is a component of $R^{M\times M}$, and that's column vectors $a_1, a_2, ..., a_M \in R^M$ are linearly independent. The result that follows is a lattice L that was created by A [36].

$$\mathsf{L} = \{x \in R^M \text{ such that } \exists s \in Z^M, x = As = \sum_{j=1}^{M} s_j a_j\}$$

Definition 1 If P is a prime number, $A_o \in Z_P^{N\times M}$ and $\eta \in Z_P^N$ [37], we have

$$\begin{aligned}
\mathsf{L}_P(A_o) =& \{v \in Z^M \text{ such that } \exists \eta \in Z_P^M \text{ where} \\
& A_o\eta = v(\text{mod P})\}, \\
\mathsf{L}_P^{\perp}(A_o) =& \{v \in Z^M \text{ such that } A_o v = 0(\text{mod P})\}, \\
\mathsf{L}_P^{\eta}(A_o) =& \{v \in Z^M \text{ such that } A_o v = \eta(\text{mod P})\},
\end{aligned}$$

Theorem 1 *Let positive integers N, P, and M, where $P \geq 3$ and $M \geq 5N \geq \log P$. The* PPT *algorithm TrapGen(P, N) may generate a pair $(A_o \in Z_P^{N\times M}, T_{A_o} \in Z^{M\times M})$ where A_o is almost uniformly distributed on $Z_P^{N\times M}$ and T_{A_o} is a basis of $\mathsf{L}_P^{\perp}(A_o)$ such that the $||T_{A_o}|| \leq M \cdot \omega(\sqrt{\log M})$ with a probability of at least $N^{\omega(1)}$* [38].

Theorem 2 *Let N, P, M, t be positive integers with $P \geq 3$ and $M \geq 2N \log P$. There exist a* PPT *algorithm SamplePre such that on input of $B \in Z^{N\times tM}$, a basis A_σ, a vector $\mathsf{L}_P^{\perp}(A_\sigma)$, and an integer $\flat \geq ||\tilde{A}_\sigma|| \cdot \sqrt{tM} \cdot w(\sqrt{\log tM})$, the distribution of the output of $T_A \leftarrow SampleBasis(B, A_\sigma, \sigma, \flat)$ is within negligible statistical distance of $||\tilde{T}_A|| \leq \flat$* [37, 39].

2.2 Complexity in lattices

The shortest integer solution (SIS) and Learning with error (LWE) problems, which are defined as follows, provide the foundation of our construction for the identity-based encryption(IBE) scheme.

Definition 2 (**SIS problem**) Given the parameters N, M, P, α and a random matrix $A_o \in Z_P^{M\times N}$, the SIS problem is solve for nonzero vector $d \in Z_P^M$ such that $||d|| \leq \alpha$ and $A_o d = 0(\text{mod P})$.

Definition 3 Assume that N is a positive integer, P is a prime number, and D is a distribution over Z_P. A (Z_P, N, D)-LWE problem instance has admittance to a challenge oracle O that isn't authorised. This challenge oracle O is either a noisy pseudo-random sampler O_S tied to a secret key $S \in Z_P^N$ or a really random sampler O_n, and they either exhibit the following characteristics:

- O_S: returns samples of the form $(a_j, b_j) = (a_j, b_j \cdot S + \phi_j) \in Z_P^N \times Z_P$, where $\phi_j \in Z_P$ is a noise sample from D and $S \in Z_P^N$ is a secret key generated at random.
- O_r: returns truly random samples from $Z_P^N \times Z_P$.

3 Problem approach

IBE enables ciphertext verification and inhibits unauthorized senders so that only legitimate decryption keys may be used to decrypt and decode the communication. The IBE system model, syntax, and security definition are provided in this section.

3.1 System model and workflow

The technology we present offers a novel approach to facilitate the secure exchange of encrypted data inside the Internet of Things (IoT) ecosystem. Figure 2 presents a comprehensive depiction of the architecture used for the purpose of exchanging data. This section will provide a comprehensive explanation of the participating entities and the workflow design of the data-sharing architecture.

As shown in Fig. 2, the system mainly comprises four entities, namely data owners (*DO*), data users (*DU*), storage service providers (*SSP*), and the blockchain network.

- **Key generation center (KGC):** The third party is responsible for generating parameters and keys for the user.
- **Data owners (DO):** The data produced by IoT devices is often of significant magnitude. IoT devices should encrypt the data they gather before sending it to third-party servers for storage and sharing. People have extensively embraced cloud computing and edge computing to store and disseminate IoT data. The distributed file system is an appropriate storage solution for managing large volumes of IoT data, including the "Interplanetary File System (IPFS)". Furthermore, the integration of distributed storage service providers with a decentralized blockchain network has the potential to enhance the security and dependability of encrypted data storage and sharing. The storage service providers in our system are responsible for keeping the encrypted data generated by IoT devices. During the process of transferring data, the secure, sensitive information protocol is unable to access or intercept the unencrypted data. Our scheme proposes a method to prevent privacy exposure to harmful entities by establishing a separation between data access grant and authentication from the data storage service.

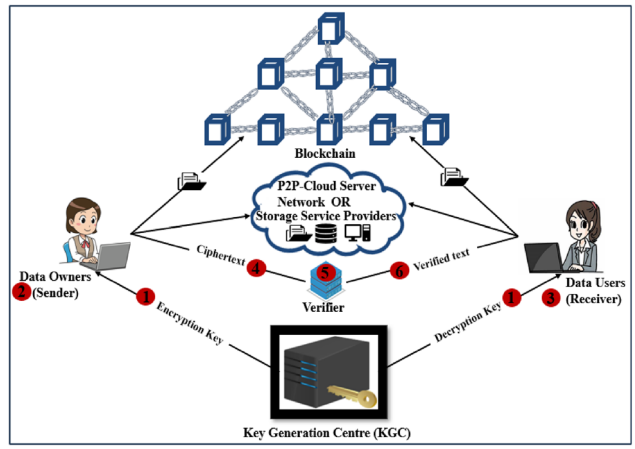


Fig. 2 System Architecture

- **Data users (DU):** The use of IoT device data is essential for a multitude of institutions and applications. Users can locate desired material by perusing the blockchain's data summary. In the preceding research work, a smart contract architecture was developed, using attribute-based access control (ABAC) and the searchable encryption algorithm [40]. Within this architectural framework, the use of blockchain extends beyond its conventional function in access control, enabling accurate and intricate access control mechanisms. Furthermore, it has the capability to accomplish efficient and privacy-protected data retrieval. *DU* has the capability to conduct searches for encrypted keywords that are kept on the blockchain. We perform these searches to retrieve the associated information. Since the primary focus of this work is not on the access control mechanism, it will not be extensively elaborated upon. After receiving consent from the data owners, the blockchain node builds an encrypted metadata ciphertext. The *DU* may then use their private key to decode the encrypted ciphertext and access the metadata information. In order to collect encrypted data, the *DU* initiates access requests to the *SSP* based on the data storage location. We can retrieve the original data by decrypting the encrypted data using the symmetric key. Moreover, the verification of a data integrity and accuracy may be achieved by the use of data summary and signature.
- **Blockchain network:** The blockchain network plays a crucial role in the functioning of this work. The

dissemination and exchange of encrypted symmetric key ciphertext relies on the encryption technique, with proxy servers playing a crucial role. Within our framework, the blockchain consensus node assumes the role of proxy servers. We implement an encryption method to maximize the benefits of the decentralized nature of the blockchain network. The designated authority distributes various shares of encryption keys to numerous nodes in the blockchain network. The consensus nodes use the matching encryption share to encrypt the original ciphertext. The metadata ciphertexts undergo transformation and consolidation into new blocks. This technique ensures that the network does not disclose or expose the confidential key belonging to the data owners or the encrypted content. The successful execution of this procedure is contingent upon the blockchain consensus algorithm that has been devised, the comprehensive elucidation of which can be found in the coming sections of this scholarly document. The decryption unit *DU* has the capability to extract the associated metadata from the encrypted ciphertext stored in the blockchain ledger. Our proposed scheme involves substituting the third-party semi-honest agent in the encryption process with a decentralized Peer-to-Peer (P2P) cloud server blockchain network. This modification allows for the establishment of a reliable and protected means of exchanging encrypted data. The blockchain ledger ensures the immutable recording of all ciphertext conversion records and data-sharing activities. Based on the available data, it is possible to readily detect the malicious access behaviors of certain proxy nodes, hence enhancing the overall security of the system.

- **Storage service providers (SSP):** The data produced by IoT devices is often of significant magnitude. IoT devices should encrypt the data they gather before sending it to third-party servers for storage and sharing. People have extensively used cloud computing and edge computing to store and disseminate IoT data. The distributed file system is an appropriate storage solution for large-scale IoT data, such as IPFS. Furthermore, the integration of distributed storage service providers with a decentralized blockchain network has the potential to enhance the security and dependability of encrypted data storage and sharing. Within our system, the SSP is responsible for storing the encrypted data from IoT devices. Throughout the data transfer process, the secure, sensitive data protocol cannot intercept or access the unencrypted information. The framework we have developed offers a solution to mitigate the risk of privacy exposure to harmful entities by creating a separation between the grant and authentication of data access and the data storage service.

3.2 Syntax

Six polynomial algorithms are included in an IBE: Setup, SK-Gen, RK-Gen, Enc, Verify, and Dec. The following is the official definition of IBE.

- *Setup* (1^ℓ): The IBE scheme is initialised by the KGC, which is considered as a reliable authority. A security parameter ℓ is used as an input while running the setup algorithm, and the outputs are the public parameter *Pk* and the master secret key *msk*. For simplicity, the common input *Pk* is left out in other algorithms.
- *Sk-Gen* (msk, ID_s): The KGC executes the Sk-Gen algorithm using the *msk* and the sender's $ID_s \in \{0,1\}^*$. The encryption key Ek_s is then returned.
- *Rk-Gen* (msk, ID_r): The Rk-Gen algorithm is executed by the KGC and requires two inputs: a receiver's $ID_r \in \{0,1\}^*$, and a *msk*. Then, a decryption key Dk_r is returned.
- *Enc* $(Ek_s, \mu, ID_r,)$: The sender executes Enc algorithm. It uses the encryption key $(Ek_s$ and the receiver's target identity ID_r to encrypt a message $\mu \in \{0,1\}$ ahead of generating a ciphertext $\mathcal{CT}$.
- *Verify*$(\mathcal{CT}, ID_s)$: The routers employs the Verify algorithm. It enters ciphertext $\mathcal{CT}$ along with the sender's and the target's identities. If and only if ID_s, it then returns a bit 1; otherwise, it returns a bit 0.
- *Dec*$(\mathcal{CT}, Dk_r)$: The receiver operates the Dec algorithm. The ciphertext $\mathcal{CT}$ is decrypted using a decryption key Dk_r. Then, if and only if ID_r, it prints a plaintext μ; otherwise, it returns $\perp$.

3.3 Security definition

We give two security definitions in our IBE scheme. They are existential unforgeability under a selected message attack $(\mathcal{EU}-\mathcal{CMA})$ and indistinguishability under a chosen-plaintext attack $(\mathcal{IND}-\mathcal{CPA})$ in the random oracle model [41].

Definition 4 $(\mathcal{IND}-\mathcal{CPA})$ The $(\mathcal{IND}-\mathcal{CPA})$ security for a IBE scheme means that the advantage

$$\mathcal{A}dv_{\mathcal{A}}^{(\mathcal{IND}-\mathcal{CPA})}(1^\ell) = |Pr[\mathcal{E}xp_{\mathcal{A}}^{(\mathcal{IND}-\mathcal{CPA})}(1^\ell)] - 1/2|$$

is negligible for any $\mathcal{PPT}$ adversary $\mathcal{A}$.

Definition 5 $((\mathcal{EU}-\mathcal{CMA}))$ The $(\mathcal{EU}-\mathcal{CMA})$ security for a IBE scheme means that the advantage

$$\mathcal{A}dv_{\mathcal{A}}^{(\mathcal{EU}-\mathcal{CMA})}(1^\ell) = Pr[\mathcal{E}xp_{\mathcal{A}}^{(\mathcal{EU}-\mathcal{CMA})}(1^\ell) = 1]$$

is negligible for any $\mathcal{PPT}$ adversary $\mathcal{A}$.

4 Proposed scheme

We are now going to give you the concrete construction scheme for IBE as well as correctness.

4.1 Concrete construction of IBE

1. **Setup** (1^{ℓ}): If N and M are positive integers and$P$ is a prime number, then they satisfy the conditions that $P \geq 3$ and $M \geq 9N \log P$. Let $H_1 : \{0,1\}^* \rightarrow Z_P^{N \times M}$, $H_2 : \{0,1\} \rightarrow Z_P^N$ and input a ℓ security parameter. After that
   (a) The KGC creates a random matrix $A_o \in Z_P^{N \times M}$ with a short basis T_{A_o} for $\mathcal{L}_P^{\perp}(A_o)$ in such a way that $||\tilde{T_{A_o}}|| \leq o(\sqrt{N \log P})$ by performing the TrapGen(P, N) method.
   (b) The master secret key is defined by the KGC as $msk = T_{A_o}$ and the public parameter is defined as $Pk = A_o$.
2. **Sk-Gen** (msk, ID_s): Using the sender's ID_s and the msk, perform the following:
   (a) The SampleBasis$(A_o|ID_s, T_{A_o}, s)$ algorithm is executed by the KGC to produce the matrix $\mathcal{B}_1 \in Z^{M \times M}$.
   (b) Return back the $Ek_s = \mathcal{B}_1$ encryption key.
3. **Rk-Gen** (msk, ID_r): Using the receiver's ID_r and the msk, perform the following:
   (a) The SampleBasis$(A_o, T_{A_o}, H_1(ID_r), s)$ algorithm is executed by the KGC to produce the matrix $\mathcal{E}_1$. Where $\mathcal{E}_1 \in Z_P^M$ satisfies $A_o\mathcal{E}_1 = H_1(ID_r) \in Z_P^N$.
   (b) Return back the $Dk_s = \mathcal{E}_1$ encryption key.
4. **Enc** $(Ek_s, \mu, ID_r,)$: Using the encryption key $\mathcal{B}_1$ of the sender, the receiver's target identification $ID_r \in \{0,1\}^*$, and the plaintext $\mu \in \{0,1\}^*$ perform the following:
   (a) Choose a random vector $v \in Z_P^N$.
   (b) Choose noise vectors $a \leftarrow D_{Z_P^N,s}$, and $b \leftarrow D_{Z_P^N,s}$.
   (c) Set $C_1 = A_o^T v + b \in Z_P^M$, $C_2 = H_1(ID_r)^T v + a + \mu\lceil P/2 \rceil \in Z_P$.
   (d) Operate algorithm SamplePre$(A_o|ID_s, \mathcal{B}_1, H_2(C_1, C_2), s) \rightarrow \beta$, where $\beta \in Z_P^{2M}$ satisfies $(A_o|ID_s)\beta = H_2(C_1, C_2) \in Z_P^N$.
   (e) Outcome of the ciphertext $\mathcal{CT} = (C_1, C_2, \beta)$.
5. **Verify**$(\mathcal{CT}, ID_s)$: Upon receiving input ciphertext $\mathcal{CT} = (C_1, C_2, \beta)$, a target identification is sent as $ID_s \in \{0,1\}^*$, do:
   (a) Check $(A_o|ID_s)\beta \stackrel{?}{=} H_2(C_1, C_2) \in Z_P^N$.
   (b) Returns 1 if the equation holds; if not, communication is stopped.
6. **Dec**$(\mathcal{CT}, Dk_r)$: Execute the following operations on input ciphertext $\mathcal{CT} = (C_1, C_2, \beta)$, a decryption key $Dk_s = \mathcal{E}_1$, and a target $Dk_r \in \{0,1\}^*$:
   (a) Evaluate $\rho = C_2 - Dk_r^T C_1$.
   (b) Analyze ρ and $\lfloor P/2 \rfloor$ in Z_P. If $|\rho - \lfloor P/2 \rfloor| < \lfloor P/4 \rfloor$, then outputs 1, else outcome 0.

4.2 Correctness

If ID_r, ID_s and $\mathcal{CT}$ is a valid ciphertext, then

$$\begin{aligned}\rho =& C_2 - Dk_r^T C_1 \\ =& H_1(ID_r)^T v + a + m\lfloor P/2 \rfloor - \mathcal{E}_1^T(A_o^T v + b) \\ =& H_1(ID_r)^T v + a + m\lfloor P/2 \rfloor - H_1(ID_r)^T v - \mathcal{E}_1^T b \\ =& m\lfloor P/2 \rfloor + a - \mathcal{E}_1^T b\end{aligned}$$

The error term is represented by the formula $x - \mathcal{E}_1^T b$. The TrapGen algorithm performs extremely well (i.e., $M > 9N \log P$), is large enough for the SampleBasis and SamplePre algorithms, and Regev's reduction [42] applies (i.e., $P > 3\sqrt{N/\eta}$), all of which are necessary for the system to function correctly.

4.3 Block formation, validation and addition in blockchain network

During this stage, we focus on the procedures concerning the formation of blocks, the validation of blocks through smart contracts, and the establishment of consensus through voting for the inclusion of these blocks.

4.3.1 Block formation

Any node acting as the sender can initiate a transaction within the P2P cloud server blockchain network, and the transaction is signed using its private key (encryption key). A node in cloud servers equipped with mining capabilities can introduce a new block in the proposed blockchain network. This node's responsibility includes gathering all transactions and creating a random oracle for the generation of a new transaction block. Each transaction is created by the cloud server received from the sender, and each transaction has the sender's identity and encryption key $\mathscr{B}_1$ represented as $Tx_i = \{C_{i,1}, C_{i,2}\}_{(\mathscr{B}_1, ID_s)}$ where $i = 1, 2, 3, \ldots$. These transactions are then used by the minor cloud server to generate a block denoted as

$Block_t = \{A_0, ID_s, ID_r, \beta, Tx_i\}$. Following the generation of the block, it is broadcast to the P2P cloud server network.

4.3.2 Block verification

Once the miner node successfully mines the block, it is disseminated to the decentralized network of cloud servers for the verification process. Each cloud server begins by verifying the sender's identity and then checks whether $(A|ID_s)\beta = ?H_2(C_1, C_2) \in Z_N^P$. If true, then the consensus process starts for adding the block to the blockchain.

4.3.3 Block addition

The cloud server within the blockchain network of the proposed protocol utilizes a voting-based consensus algorithm to establish agreement regarding a specific block. The proposed system has implemented the "Practical Byzantine Fault Tolerance (PBFT)" protocol, a well-known voting-based consensus mechanism. Algorithm 1 outlines the complete series of steps involved in the consensus process, encompassing the verification of smart contracts.

Algorithm 1 Consensus Mechanism for block validation and addition in Blockchain

Input: Suppose a leader $Leader_{CS_m}$ picks a block, $Block_t = \{A_0, ID_s, ID_r, \beta, Tx_i = \{C_{i,1}, C_{i,2}\}_{(\mathscr{B}_1, ID_s)}$ where $i = 1, 2, 3..., n_t\}$
Output: Status of block commitment (YES/NO)

```
Set a Target Number = 2 * (M − 1)/3 + 1
CSN_i ← (empty)
Distribute Block_t to replica nodes in cloud servers network
for every cloud server replica node RCS_j do
    Set Vote_Consensus_j = NO
    Calculate Block_Hash = Hash(Block_t)
    if (Block_Hash = Curr_Block_h) then
        if ((A|ID_s)β =?H_2(C_1, C_2) ∈ Z_N^P) then
            Calculate Merkle tree root (MTR'_Tx_i) utilizing the n_i
            transactions contained within the block payload
            if (MTR'_Tx_i = MTR_Tx_i) then
                Set Vote_Consensus_j = YES
            end if
        end if
    end if
    Add Vote_Consensus_j to CSN_i
end for
Set Vote_Count ← 0
for each vote (Vote_Consensus_j) reply to CSN_i do
    if (Vote_Consensus_j is YES) then
        Vote_Count = Vote_Count + 1
    end if
end for
if (Vote_Count ≥ Target Number) then
    Add Block_t into the blockchain
    Share the status of block commitment as "YES" with the entire
    blockchain network
end if
```

5 Security analysis

In order to ensure the security of our suggested IBE scheme based on the LWE assumption, we now demonstrate the following theorem.

Theorem 3 *If the decisional LWE assumption is true, then our suggested IBE method with parameters (N, M, s, ζ, P) comparable to section 4.2 cannot be broken by $\mathcal{PPT}$ adversaries.*

Proof Assume IBE does not meet the $\mathcal{IND}-\mathcal{CPA}$ security definition. The suggested technique can then be broken by an adversary $\mathcal{A}$ with a non-negligible advantage ϵ. As a result, we can create algorithm $\mathbb{B}$ to solve the decisional LWE problem by interacting with the adversary $\mathcal{A}$. The challenge sends a LWE problem $(\lambda_j \in Z_P^N, \gamma_j \in Z_P)$ to $\mathbb{B}(i \le j \le M)$, where $\gamma_j = \lambda_j^T v + y_j (y_j \leftarrow D_{Z_p^N})$ or chosen at random. Algorithm $\mathbb{B}$ interacts with algorithm $\mathcal{A}$ in the following manner to solve the LWE problem.

- *Initially:* $\mathcal{A}$ presents ID_r^* as a challenge.
- *Setup:* $\mathbb{B}$ puts together $A_o \in Z_P^{N\times M}$ by establishing $A_o = (\lambda_1, \lambda_2, ..., \lambda_M)$. Then $\mathbb{B}$ gives $Pk = A_o$ to $\mathcal{A}$.
- *Hash Queries:* In the possibilities, $\mathbb{B}$ uses the algorithm TrapGen to generate $A_j \in Z_P^{N\times M}$ and a short basis $G_j \in Z_P^M$ for lattice $\mathcal{L}_P^{\perp}(A_j)$, where $A_j \in Z_P^{N\times M}$ random, adds ID_s, A_j, G_j to L_1, and then sends A_j to the adversary $\mathcal{A}$. $\mathbb{B}$ chooses at random $P^* \in [1, P_{H_1}]$. If the P_j^{th} query ID_r appears in the list $P_j, ID_r, \lambda_j, \mathcal{E}_j$, then L_2 is true. $\mathbb{B}$ gives $\mathcal{A}$ the result $H_(P_j) = \lambda_j$. If $P_j = P^*$, then $\mathbb{B}$ determines $\lambda_j = \lambda^*$ and picks $\mathcal{E}_j \in Z_P^{2M}$ at random. Alternatively, $\mathcal{B}$ adds $P_j, ID_r, \lambda_j, \mathcal{E}_j$, to L_2, samples $\mathcal{E}_j$ from the distribution $\mathcal{D}_{Z^M,s}$ such that $A_o\mathcal{E}_j = \lambda_j$, and returns λ_j to the adversary $\mathcal{A}$.

Query phase 1:

- *Sk-Gen queries:* An adversary gives ID_s. If the values $\{ID_s, A_j, G_j\}$ are not found in the list L_1, algorithm $\mathbb{B}$ then asks the oracle hash for the values $\{ID_s, A_j, G_j\}$. Otherwise, $\mathbb{B}$ generates a matrix $\mathcal{B}_j Z$ using the procedure SampleBasis$(A_o|A_i, G_i, ID_s, s)$ and gives the adversary $\mathcal{A}$ the encryption key $Ek_s = \mathcal{B}_j$.
- *Rk-Gen queries:* The adversary $\mathcal{A}$ offers ID_r. If the values $P_j, ID_r, \lambda_j, \mathcal{E}_j$ are not found in the list L_2, algorithm $\mathbb{B}$ then asks the oracle hash for the values $P_j, ID_r, \lambda_j, \mathcal{E}_j$. When $P_j = P^*$, $\mathbb{B}$ fails. If not, $\mathbb{B}$ gives the adversary $\mathcal{A}$ the decryption key $Dk_r = \mathcal{E}_j$.

Challenge: The adversary $\mathcal{A}$ chooses two $ID_{s_1}, ID_{s_2} \in \{0,1\}^*$ and two plaintexts $\mu_1, \mu_2 \in \{0,1\}^*$with the restriction that in Query phase 1, ID_{s_1}, ID_{s_2} have never been queried. $\mathbb{B}$ first selects $i \in \{0,1\}$ before requesting information from Oracle hash and receiving ID_s^*, A_i^*, G_i^*. After

that, $\mathbb{B}$ generates a matrix $\mathcal{B}_i \in Z_P^{M\times N}$ using the procedure SampleBasis$(A_o|A_i^*, G_i^*, s)$ and the encryption key is $Ek_s = \mathcal{B}_i^*$. $\mathbb{B}$ terminates if $P_j \neq P^*$, else $\mathbb{B}$ specifies the ciphertext $\mathcal{CT}^*$ as follows:

1. Set $\gamma^* = (\gamma_1, \gamma_2, ..., \gamma_M)^T \in Z_P^M$.
2. A plaintext μ_i can be encrypted by setting $C_1^* = \gamma_1 + \mu_i \lfloor P/2 \rfloor \in Z_P^M$.
3. Execute the algorithm SamplePre $(A_o|A_i^*, \mathcal{B}_i^*, H_3(\gamma^*, C_1^*))$, to generate a vector $\mu^* \in Z_P^M$.
4. Send $\mathcal{CT}^* = (\gamma^*, C_1^*, \beta^*)$ to the adversary $\mathcal{A}$.

Query phase 2: In the Query Phase 1, it is stated that the adversary $\mathcal{A}$ can query the method $\mathbb{B}$ to obtain the encryption key and decryption key, but that $ID_{s_i} \neq ID_{s_0}$ and ID_s.

Guess: $\mathcal{A}$ produces the bit i^*. If $i^* = i$, then $(\lambda_j, \gamma_j) \in Z_P^N \times Z_P$ belongs to the distribution $\mathcal{O}_S$. If not, $(\lambda_j, \gamma_j) \in Z_P^N \times Z_P$ is evenly sampled from $Z_P^N \times Z_P$.

Let's now examine the probability of a successful simulation. In the first query phase and the challenge phase, the game's termination probabilities are $1/P_{H_1}$ and $1 - 1/P_{H_1}$, respectively. Therefore, the probability that the simulation would succeed is $1 - 1/P_{H_2}(1 - 1/P_{H_2})$. If $(\lambda_j, \gamma_j) \in Z_P^N \times Z_P$ belongs to the distribution $\mathcal{O}_S$, then i = u T i v + yi (yi) is exactly we have $\gamma_j = \lambda_j^T v + b_j$. As a result, the ciphertext $\mathcal{CT} = (\gamma^*, C_2^*, \beta^*)$ created in the Challenge phase is valid

$$\begin{aligned}
\gamma^* =& (\gamma_1, \gamma_2, ..., \gamma_M)^T \\
=& A_o^T v + b, \\
C_2^* =& \gamma_1 + \mu_i \lfloor P/2 \rfloor, \\
(A_o|A_i^*)\beta^* =& H_2(\gamma^*, C_2^*)
\end{aligned}$$

and $\mathcal{CT}^*$ is valid challenge ciphertext. As a result, adversary $\mathcal{A}$ is ϵ an advantage, and $|Pr[i^* = i]| \geq 1/2 + \epsilon$. If $(\lambda_j, \gamma_j) \in Z_P^N \times Z_P$ is uniformly sampled from $Z_P^N \times Z_P$ then γ_j is random in Z_P, which then results in C_2^* being random in Z_P. As a result, no legitimate adversary can learn anything about $i \in \{0, 1\}$ from the challenge ciphertext $\mathcal{CT}^*$. Hence, $|Pr[i^* = i]| \geq 1/2 + \epsilon$. As a result, $\mathbb{B}$ has an advantage in disproving the LWE supposition. □

Theorem 4 *If the SIS assumption is correct, our suggested IBE technique with parameters (N, M, s, ζ, P) comparable to section 4.2 cannot be compromised by PPT adversaries.*

Proof Theorem 3's proof is similar to this one; please see the full version of it. □

5.1 SCYTHER tool

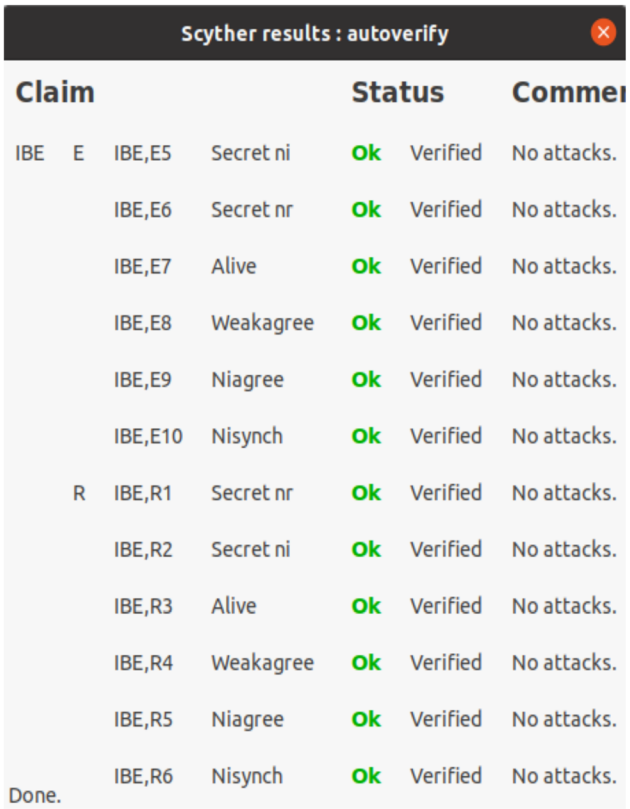
Currently, there are multiple tools for conducting formal security analysis, including TAMARIN [43], AVISPA [44], Proverif [45], CryptoVerif [46], and Scyther [47]. In this analysis, we evaluate the security of our proposed scheme by utilizing the Scyther tool. The Scyther is a tool designed to assess the security features of a protocol by evaluating its security claims. The Scyther tool accepts a description of the protocol in the "security protocol description language (SPDL)" structure as input. It has the capability to establish security claims for each role in the protocol and verify the accuracy of the claims produced. If the claim is inaccurate, its attack structure is visually represented. The proposed technique was implemented in SPDL, and the result has been verified by "ok". The Scyther tool successfully verified the proposed scheme's automatic security verification results, as depicted in Fig. 3.

6 Performance evolution

In this portion, a comparison is shown between the proposed system and existing schemes. These schemes are all based on multi-authority attribute based encryption on lattices. In order to provide an equitable evaluation, the parameters for all schemes are set to identical values, as seen in Table 2.

6.1 Computation cost

The cost estimation is conducted using the realistic characteristics provided in Table 2. The operations were implemented using Python version 3.11.6, with the Numpy library version 1.24.3. The experiments were conducted on a Windows 11 Enterprise machine, equipped with an Intel(R) Core(TM) i7-7700 CPU running at 3.60GHz and 16.00GB of RAM. The notations and computation costs of the operators are provided in Table 3, while the computation costs for encryption and decryption are found in Table 4. Moreover, a visual representation of the comparison can be seen in Fig. 4 and Fig. 5. We can see that the proposed scheme's encryption cost is $T_{*1} + (T_{*2} + T_{*3} + T_{+1}) \approx 3.096$ and the decryption cost is $T_{*6} \approx 0.010$. Therefore, with a significant reduction in execution time, the performance evaluation along with the comparative analysis demonstrates the validity of the proposed lightweight scheme as it mitigates the issue of the NMID's computational capability.



Scyther results : autoverify

Claim				Status		Commer
IBE	E	IBE,E5	Secret ni	Ok	Verified	No attacks.
		IBE,E6	Secret nr	Ok	Verified	No attacks.
		IBE,E7	Alive	Ok	Verified	No attacks.
		IBE,E8	Weakagree	Ok	Verified	No attacks.
		IBE,E9	Niagree	Ok	Verified	No attacks.
		IBE,E10	Nisynch	Ok	Verified	No attacks.
	R	IBE,R1	Secret nr	Ok	Verified	No attacks.
		IBE,R2	Secret ni	Ok	Verified	No attacks.
		IBE,R3	Alive	Ok	Verified	No attacks.
		IBE,R4	Weakagree	Ok	Verified	No attacks.
		IBE,R5	Niagree	Ok	Verified	No attacks.
		IBE,R6	Nisynch	Ok	Verified	No attacks.

Done.

Fig. 3 The security verification results of the IBE scheme in the Scyther tool

6.2 Communication cost

The proposed protocol undergoes a comparison with the state-of-the-art approaches concerning communication costs for the encryption and decryption phases. To account for the communication cost of pertinent protocols, the bit sizes considered for all parameters are displayed in Table 2. The variation in communication cost for public parameter size,encryption key size, ciphertext size, and decryption key size phases is given in Table 5.

Similarly, Table 5 depicts the comparison of the storage and communication costs for the public parameter size, encryption key size, ciphertext Ciphertext Size and Decryption size. Lucidly, it can be observed from Table 5 that the proposed scheme offers better and lower communication costs than the other protocols. Therefore, we can deduce from the preceding information that the suggested scheme outperforms other schemes in terms of efficiency and performance.

6.3 Feature comparison

The proposed scheme is evaluated with some existing protocols on the premises of various security features in Table 6. Considered parameter of the comparison are following: a) Quantum resistance, b) Multi-authority, c) SCYTHER Tool, and d) Security Proof. Lucidly, it can be depicted from the comparison shown in Table 6 that the proposed mechanism provides a number of advantages over the pre-existing protocols.

7 Conclusion

Blockchain's hack-proof encryption eliminates IoT security risks. This technology can also track connected sensors and nodes. A secure lattice based encryption scheme for IoT-oriented network computing systems based on blockchain helps securely offload data from devices. We conducted several tests to ensure that our proposed algorithm is secure. However, there are still some limitations to using this technology, such as limited computing resources and

Table 2 Parameters

Sr. N	Security Parameters (N)	Basis dimension (M)	Modulo (P)	Total number of users (U)	Size of attribute set ($\mathbb{H}$)	Total number of attribute authorities ($\mathbb{K}$)
1	2	1024	2^2	10	10	5
2	4	2048	2^4	50	10	5
3	8	2048	2^8	50	10	5
4	16	2048	2^{16}	50	10	5
5	32	4096	2^{32}	100	10	5
6	64	4096	2^{64}	100	10	5
7	128	4096	2^{128}	100	10	5
8	256	4096	2^{256}	100	10	5
9	512	4096	2^{512}	100	10	5
10	1024	4096	2^{1024}	100	10	5

Table 3 The notations and operation cost

Sr. N	Notation	Description	Cost (*ms*)
1	T_{*1}	The computational time of multiplication $\mathcal{Z}_P^{1\times N} \times \mathcal{Z}_P^{1\times N}$	0.002
2	T_{*2}	The computational time of multiplication $\mathcal{Z}_P^{M\times N} \times \mathcal{Z}_P^{1\times N}$	3.052
3	T_{*3}	The computational time of multiplication $\mathcal{Z} \times \mathcal{Z}_P^{1\times N}$	0.002
4	T_{*4}	The computational time of multiplication $\mathcal{Z} \times \mathcal{Z}_P^{1\times M}$	0.082
5	T_{*5}	The computational time of multiplication $\mathcal{Z}_P^{M\times N} \times \mathcal{Z}_P^{1\times M}$	2.569
6	T_{*6}	The computational time of multiplication $\mathcal{Z}_P^{1\times M} \times \mathcal{Z}_P^{1\times M}$	0.010
7	T_{+1}	The computational time of an addition $\mathcal{Z}_P^{1\times N} + \mathcal{Z}_P^{1\times N}$	0.040

Table 4 Computation cost

Sr. N	Schemes	Encryption Cost (*ms*)	Decryption Cost (*ms*)
1	[33]	$T_{*1} + \mathbb{H} \times (T_{*2} + T_{*3} + T_{+1}) \approx 30.942$	$\mathbb{H} \times (T_{*4} + T_{*5}) + \mathbb{K} \times T_{*4} + \mathbb{H} \times T_{*6} \approx 27.02$
2	[48]	$T_{*1} + \mathbb{H} \times (T_{*2} + T_{*3} + T_{+1}) \approx 30.942$	$\mathbb{H} \times T_{*6} \approx 0.10$
3	[35]	$\mathbb{H} \times T_{*2} + \mathbb{K} \times \mathbb{H} \times (T_{*2} + T_{+1}) \approx 185.12$	$\mathbb{K} \times T_{*3} + 2\mathbb{K} \times T_{*6} + \mathbb{K} \times T_{+1} \approx 0.31$
4	[34]	$T_{*1} + \mathbb{H} \times (T_{*2} + T_{*3} + T_{*5} + T_{+1}) \approx 56.632$	$\mathbb{H} \times (T_{*4} + T_{*5}) + \mathbb{H} \times T_{*4} + \mathbb{K} \times T_{*4} + \mathbb{H} \times T_{*6} \approx 27.84$
5	[38]	$T_{*1} + (T_{*2} + T_{*3} + T_{*5} + T_{+1}) \approx 5.665$	$T_{*6} + T_{*5} \approx 2.579$
6	[49]	$T_{*1} + \mathbb{H} \times (T_{*2} + T_{*3} + T_{*5} + T_{+1}) \approx 56.632$	$\mathbb{H}(T_{*6} + T_{*5}) \approx 25.79$
7	Proposed	$T_{*1} + (T_{*2} + T_{*3} + T_{+1}) \approx 3.096$	$T_{*6} \approx 0.010$
Proposed Scheme	$U = 10$	$U = 50$	$U = 100$
Encryption cost	30.96 *ms*	154.8*ms*	309.6*ms*

$\mathbb{H}$: Represents the size of the attribute set associated with the ciphertext, U: Represents the total numbers of users, $\mathbb{K}$: Represents the total number of attribute authorities

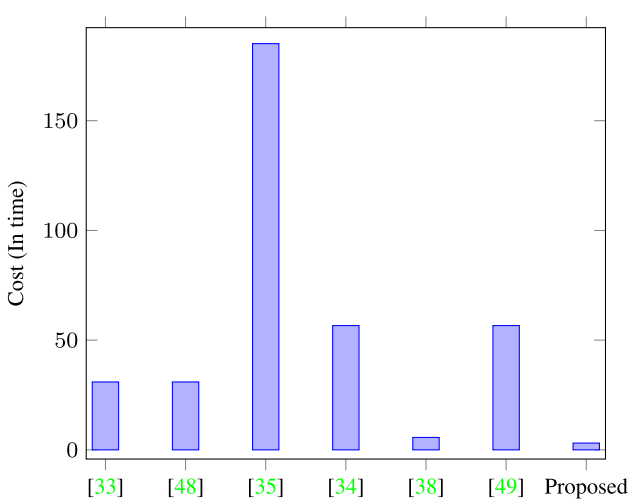


Fig. 4 Comparison based on encryption cost

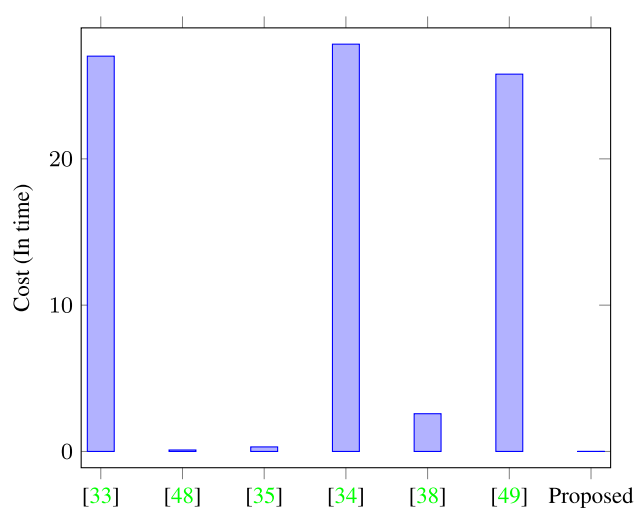


Fig. 5 Comparison based on decryption cost

transaction speed. Many IoT devices, such as connected sensors, lack memory and processing resources and cannot act as nodes in a blockchain. Web services can solve this problem, but this problem still needs to be fixed. In the future, security will also be considered for data in the cloud after offloading. Further research is underway to make this technology more efficient and adaptable.

Although our protocol is efficient in a single cloud and single-user environment, we want to expand our work in the future to provide batch verification in multi-cloud and

Table 5 Communication costs for state-of art

Sr. N	Schemes	Public Parameter Size	Encryption Key Size	Ciphertext Size	Decryption Key Size
1	[33]	$(3NM+N)\|Z_P\|$	-	$2M\|Z_P\|$	$(2M+1)\|Z_P\|$
2	[48]	$(3NM+N)\|Z_P\|$	-	$2M\|Z_P\|$	$(2M+1)\|Z_P\|$
3	[35]	$(NM+N^2)\|Z_P\|$	-	$2M\|Z_P\|$	$(3M+1)\|Z_P\|$
4	[34]	$(NM+N)\|Z_P\|$	-	$M\|Z_P\|$	$(2M+1)\|Z_P\|$
5	[38]	$MN\|Z_P\|$	$M^2\|Z_P\|$	$NM\|Z_P\|$	$(3M+1)\|Z_P\|$
6	[49]	$MN\|Z_P\|$	$M^2\|Z_P\|$	$2M\|Z_P\|$	$(M+N)\|Z_P\|$
7	Proposed	$MN\|Z_P\|$	$M^2\|Z_P\|$	$M\|Z_P\|$	$(2M+1)\|Z_P\|$

Table 6 Properties comparison

Sr. N	Schemes	Quantum resistance	Multi-authority	Collusion Resistance	SCYTHER Tool	Security Proof
1	[33]	✓	✓	×	×	$\mathcal{CPA}$
2	[48]	✓	✓	✓	×	$\mathcal{CPA}$
3	[35]	✓	✓	✓	×	$\mathcal{CPA}$
4	[34]	✓	✓	✓	×	$\mathcal{CPA}$
5	[38]	✓	×	✓	×	$\mathcal{CCA}$
6	[49]	×	×	×	×	$\mathcal{CCA}$
7	Proposed	✓	✓	✓	✓	$\mathcal{CPA}$

$\mathcal{CCA}$: Chosen-Ciphertext Attack, $\mathcal{CPA}$: Chosen-Plaintext Attack

multiverse environments in order to improve the propose protocol's effectiveness further. Although the suggested protocol only allows static data verification; however, our work could be expanded to support dynamic data manipulations as well. We would also like to set up a complete test bed experiment in the future. With the benefits of blockchain technology, a decentralized audit mechanism can be developed that will completely eliminate third parties from the model.

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Declarations

Conflict of interest The authors declare no Conflict of interest.

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Sunil Prajapat received the M.Sc. degree in mathematics from the Central University of Himachal Pradesh, Dharamshala, India, in 2021, where he is currently pursuing the Ph.D. degree with the Srinivasa Ramanujan Department of Mathematics. His research interests are quantum cryptography, post-quantum cryptography, coding theory, Blockchain, and various applications of cryptographic primitives in the real world. He is a renowned Reviewer for numerous IEEE, Taylor and Francis, MDPI, PLOS One, Elsevier, and Springer journals. Mr. Prajapat is a CSIR Fellow.

Neeraj Kumar received the M.Sc. degree in mathematics from Motilal Nehru National Institute of Technology Allahabad, Uttar Pradesh, India. Currently, he is pursuing PhD from the department of Mathematics and Scientific Computing, National Institute of Technology Hamirpur, Himachal Pradesh, India. His research area is Cryptography and Blockchain Technology.

Ashok Kumar Das received a Ph.D. degree in computer science and engineering, an M.Tech. degree in computer science and data processing, and an M.Sc. degree in mathematics from IIT Kharagpur, India. He is currently a full professor with the Center for Security, Theory and Algorithmic Research, IIIT, Hyderabad, India. He is an adjunct professor at the Korea University, Seoul, South Korea. He also worked as a visiting research professor with the Virginia Modeling, Analysis, and Simulation Center, Old Dominion University, Suffolk, VA 23435, USA. His current research interests include cryptography, system and network security, including security in smart grids, Internet of Things (IoT), Internet of Drones (IoD), Internet of Vehicles (IoV), Cyber-Physical Systems (CPS), and cloud computing, intrusion detection, Blockchain, AI/ML security, and post-quantum cryptography. He has authored over 435 papers in international journals and conferences in the above areas, including over 370 reputed journal papers. He was a recipient of the Institute Silver Medal from IIT Kharagpur. He has been listed in the Web of Science (ClarivateTM) Highly Cited Researcher 2022 and 2023 in recognition of his exceptional research performance. He was/is on the editorial board of IEEE Transactions on Information Forensics and Security, IEEE Systems Journal, Journal of Network and Computer Applications (Elsevier), Computer Communications (Elsevier), Journal of Cloud Computing (Springer), Cyber Security and Applications (Elsevier), IET Communications, KSII Transactions on Internet and Information Systems, and International Journal of Internet Technology and Secured Transactions (Inderscience). He also served as one of the Technical Program Committee Chairs of the first International Congress on Blockchain and Applications (BLOCKCHAIN'19), Avila, Spain, June 2019, International Conference on Applied Soft Computing and Communication Networks (ACN'20), October 2020, Chennai, India, second International Congress on Blockchain and Applications (BLOCKCHAIN'20), L'Aquila, Italy, October 2020, and International Conference on Applied Soft Computing and Communication Networks (ACN 2023), Bangalore, India, December 2023. His Google Scholar h-index is 89, and i10-index is 275, with over 23,300 citations.

Pankaj Kumar received the M.Sc. from CCS University Meerut India and Ph.D. degrees from Galgotias University in 2005 and 2020, respectively. He has been an assistant professor at Srinivasa Ramanujan Department of Mathematics in the Central University of Himachal Pradesh, Dharamshala, India. He has published over 50 academic research papers on information security and privacy preservation. His current research interests include cryptography, Blockchain, wireless network security, information theory, and network coding.

Rifaqat Ali is presently working as an Assistant Professor in the Department of Mathematics and Scientific Computing, National Institute of Technology, Hamirpur, Himachal Pradesh, India. He did his PhD from the Department of Computer Science and Engineering, Indian Institute of Technology (Indian School of Mines), Dhanbad in 2018. He has more than 5 years of teaching and research experiences, and contributed a several research papers in many reputed journals and international conference proceedings. His main research interest includes Cryptography and Network security.